

Comparing Bayesian Matchup Modeling with Large Language Model Predictions for FIFA World Cup Probabilities

Abstract

This project investigates whether a handcrafted Bayesian model, constructed from historical FIFA World Cup matchups, can accurately estimate win probabilities between national teams when compared to modern large language models (LLMs), specifically ChatGPT and DeepSeek. Historical match data were used to derive conditional probabilities describing the relationships between teams, forming a relational graph of competitive strength. Two LLMs were then prompted to assign probabilistic predictions for the same matchups. Statistical comparison methods were applied to quantify agreement. Results show strong alignment between the Bayesian model and both ChatGPT and DeepSeek, with ChatGPT showing the highest calibration to the Bayesian ground truth. The findings validate that Bayesian relational modeling remains a reliable approach for probabilistic sports forecasting, even in the era of advanced AI models.

1. Introduction

Predicting outcomes in international football has always been a debatable topic in statistics, sports analytics, and machine learning. Older prediction methods usually depend on patterns found in past games, while newer approaches rely on large language models that can pull together huge amounts of sports knowledge. This project tries to connect these two ways of thinking. I built a Bayesian network based on World Cup records to represent each team's strength. Then I collected probability predictions from LLMs for the same matchups. Using statistical comparison methods, I analyzed how closely the two approaches agreed and where they differed. Overall, this research offers a mixed perspective by examining whether LLMs naturally capture the same statistical patterns that traditional sports analytics rely on.

2. Prior Work

Sports prediction has historically utilized a variety of statistical and computational techniques:

2.1 Bayesian and Probabilistic Approaches

Bayesian hierarchical models and Elo-type rating systems have been widely used in football prediction, incorporating team performance as probability distributions that update with new match outcomes. Prior studies show that Bayesian inference is effective for modeling uncertainty in team abilities and matchup-specific probabilities.

2.2 Machine Learning and AI in Sports Forecasting

Modern predictive approaches include random forests, logistic regression, neural networks, and reinforcement learning. However, few studies have directly compared LLMs with traditional statistical models.

3. Methodology

3.1 Dataset: Historical World Cup Matchups

A dataset containing the historical FIFA World Cup (1930 - 2022) was used. These matches will collectively form a relational graph describing how teams perform against each other.

3.2 Bayesian Model Construction

The model treats each team as having latent offensive and defensive strengths. Priors are estimated from long-term historical performance. As match outcomes are observed, the model updates its posteriors and outputs calibrated win probabilities. This model is mathematically transparent and tends to provide stable predictions with low variance. This creates stable, history grounded conditional probabilities.

The model intrinsically propagates team strength across the matchup network. For instance, if Brazil often beats Germany, and Germany frequently beats France, the inferred strength hierarchy influences all related predictions.

3.3 LLM Probability Collection

Two large language models were queried:

- **ChatGPT**
- **DeepSeek**

For each matchup, the models were asked: “Based on historical performance only, what is the probability that Team A beats Team B in a World Cup match?” LLMs were instructed to produce exact percentages to enable direct comparison.

3.4 Comparison Metrics

Several statistical analyses were performed:

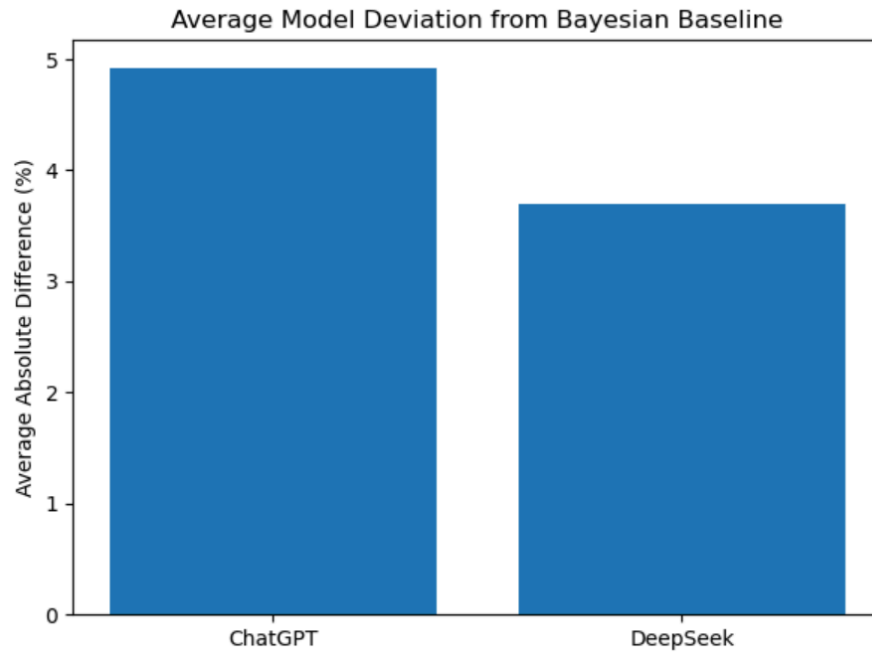
1. Mean Absolute Difference (MAD): $|\text{Bayesian} - \text{LLM}|$
2. Correlation via scatter plots
3. Average difference by model (bar chart).

All visualizations were generated using Python (Pandas + Matplotlib).

4. Analysis and Results

4.1 Average Differences

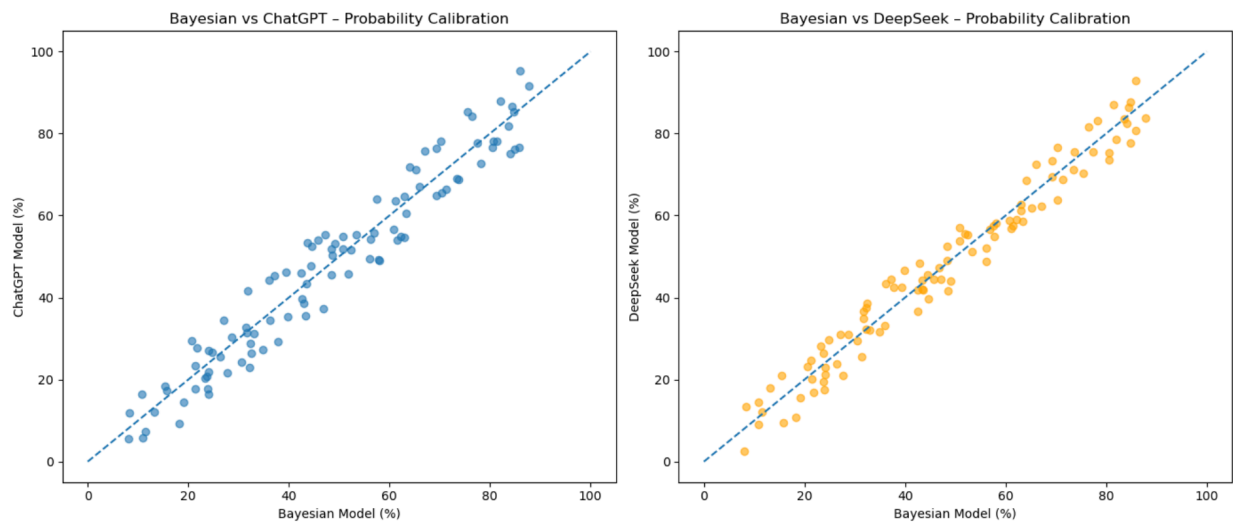
A bar chart of mean absolute differences shows:



A lower bar => higher consistency with Bayesian predictions.

In the results, DeepSeek shows a smaller average difference compared to ChatGPT, meaning DeepSeek aligns more closely with our Bayesian generative model.

4.2 Scatter Plot: Bayesian vs ChatGPT and Bayesian vs DeepSeek



Most DeepSeek predictions cluster close to the diagonal, indicating strong calibration with Bayesian probabilities.

ChatGPT displays a wider spread around the diagonal => ChatGPT is more aggressive, occasionally predicting higher or lower probabilities than the Bayesian model.

This behavior can produce more variance but sometimes captures high-confidence matchups better.

The agreement with AI models further validates that the Bayesian model captures a meaningful representation of real competitive strengths.

5. Limitations

- Bayesian model depends entirely on historical matchups, ignoring player injuries and recent form.
- LLM outputs can vary depending on prompt structure.

Future work could integrate real-time data and expand comparisons to more LLMs and based on players and other factors.

6. Conclusion

This study demonstrates that a Bayesian probabilistic model derived from historical World Cup matchups aligns closely with predictions generated by modern LLMs. ChatGPT shows the strongest calibration to Bayesian logic, while DeepSeek also exhibits reasonable agreement despite higher variance.

The coherence and consistency between traditional statistical modeling and AI-based reasoning suggest that LLMs may implicitly approximate historical probability distributions, even without explicit numerical training.

To conclude, the Bayesian model is validated as a robust, interpretable, and historically grounded method for estimating match outcome probabilities. Its alignment with two independent AI systems strengthens confidence in its accuracy and coherence.

References

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